Royal Devon University Healthcare NHS Foundation Trust

Clinical Guideline for: **Critical Airway Management: Northern Services**

SUMMARY

This guideline:

- provides a framework for anaesthetists at North Devon District Hospital to manage the critical airway (i.e. a deteriorating difficult airway with the potential for airway obstruction) without the assistance of ENT surgeons.
- identifies additional training needs to help anaesthetists manage the critical airway, including additional training in emergency front of neck access.

KEY POINTS

The main elements of this guideline are:

- A structured approach to airway management may improve outcomes (see Figure 1 overleaf). Difficult Airway Society (DAS) guidelines are available in Appendix 1.
- Successful outcomes depend on the right team being available in the right place with appropriate equipment.
- Checklists are effective in emergency / stressful situations as they reduce cognitive load and facilitate a shared mental model e.g. Appendix 2.
- The additional risk posed by critical airways at NDDH can be mitigated by additional training. Training recommendations are set out in Section 5 of this document.

Figure 1 Management of the critical airway

Presentation and assessment

Patient presents with critical airway

Assess patient

ABC
Pathology: e.g. review CT
Airway: consider nasendoscopy with AmbuScope
Assess urgency and possibility of transfer out

Prepare team

Anaesthetic consultant
ICU consultant
Contact ENT for advice - on-call at RD&E (Wonford), or at NDDH via switch
ODP
Inform theatres

Transfer

Location

Consider safety of transfer
Assistance available?
Space - consider theatre > anaesthetic room

Transfer
Theatres / ICU / Resus

Preparation

Prepare equipment

O_2 - consider high flow nasal
Bed / trolley
Pillow / positioning
Monitoring
Drugs / sedation / muscle relaxant
Difficult airway trolley
VL / bougie / stylet
CICO kit

Use Checklist

Team Roles

Team leader
Intubator: 1st, 2nd
ODP / assistant
Drugs / sedation
Monitoring / timing
Runner
eFONA

Pre-intubation

Prepare patient

Consent
Discuss possible difficulty
O_2 - consider high flow nasal
Identify cricothyroid membrane and consider airway ultrasound
Ramping / pillows etc

Airway plan

Consider awake tracheal intubation
Plan A / B / C / D
e.g. ETT / LMA / BVM / cricothryoidotomy
How many attempts at each stage?
Verbalise the plan!

Use Checklist

Aftercare

Aftercare

Liaise with ICU / ENT
Arrange transfer
Document - add to problem list and history
Datix / DAS database
Debrief

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1. INTRODUCTION

1.1 This clinical guideline is designed to provide advice, guidance and direction to staff whilst leaving room for professional judgement, and adaptation, to fit individual circumstances.

1.2 This guideline focuses specifically on the management of the critical airway where urgent intervention is required, with some time for planning, but without the assistance of ENT surgeons. It is not intended to provide guidance on management of the unexpected difficult airway. It is intended for use by anaesthetists and all other individuals at North Devon District Hospital (NDDH) who perform airway management in an emergency situation, e.g.:

- Anaesthetists (all grades)
- Anaesthetic practitioners (ODPs and anaesthetic nurses)
- Intensivists
- Emergency physicians

2. BACKGROUND

2.1 The primary goal of airway management in anaesthesia is the delivery of oxygen to a patient's lungs. Failure to deliver oxygen may rapidly result in permanent harm or death and therefore any difficulty requires prompt intervention.

2.2 Analysis of difficult airway incidents has shown that many are preventable with improved management strategies and training.[1] Nevertheless, there have been recent incidents at NDDH where it has been impossible to oxygenate patients through conventional anaesthetic means.

2.3 In emergency situations with evolving airway obstruction, attempts by anaesthetists to manage difficult airways with conventional interventions (e.g. bag valve mask ventilation, supraglottic airways or tracheal intubation) are more likely to fail, resulting in a 'Can't Intubate, Can't Oxygenate (CICO)' situation.

2.4 Such emergencies, described in this document as 'critical airways', may conventionally involve intervention by ENT surgeons (e.g. awake or emergency tracheostomy) or, in the absence of ENT, there may be a

need for urgent interventions by anaesthetists, i.e. emergency front of neck access (eFONA), involving needle or surgical cricothryoidotomy.

2.5 ENT cover is not routinely available at NDDH. This presents significant organisational risk and individual risk for anaesthetists who may be required to undertake eFONA without ENT backup.

2.6 The Difficult Airway Society has produced guidelines for the management of the difficult airway in a variety of circumstances[2-8] (see Appendix 1). Airway management in the emergency situation is addressed further by international guidelines[9-11]. This guideline incorporates evidence-based guidance from some of these documents. Universal airway guidance may become available at some stage.[12]

3. DEFINITIONS

3.1 A difficult airway is defined as 'the clinical situation in which a trained anaesthetist experiences difficulty with ventilation via the upper airway, difficulty with tracheal intubation, or both'.

3.2 A critical airway describes the situation in which a potential difficult airway is deteriorating with impending airway obstruction.

3.3 Can't intubate, can't oxygenate (CICO): this scenario arises when attempts to manage the airway by tracheal intubation, face-mask ventilation, or placement of a supraglottic airway device have failed. Hypoxic brain injury and death will result unless there is rapid resolution.

3.4 Emergency front of neck access (eFONA): the securing of a patent airway via the anterior neck to facilitate emergency alveolar oxygenation.

4. MAIN BODY OF GUIDELINE

Location

4.1 This guideline is intended to apply to anywhere difficult airways may be encountered at NDDH including, but not exclusively, in the perioperative, emergency department and critical care settings.

4.2 It is anticipated that if time permits, patients requiring difficult airway management will be transferred to a location where appropriate equipment, space and assistance is available (e.g. main theatres, the emergency department resuscitation area “resus” or critical care).

4.3 Within the theatre complex, consideration should be given to induction of anaesthesia in theatre rather than in the anaesthetic room in order to provide more space for equipment and personnel.

Preparation of the team

4.4 Whenever a difficult airway is known or suspected the on-call consultant anaesthetist should be contacted. Critical airways require the input of consultant anaesthetists and the on-call intensivist. An anaesthetic practitioner (ODP) should be contacted for assistance. Early involvement of senior specialists allows planning and a structured approach to the difficult airway.

4.5 ENT surgeons at The Royal Devon and Exeter Hospital (Wonford) may be contacted for advice and must be contacted if an ENT related transfer is anticipated. There are occasions when ENT surgeons are present at NDDH or nearby and attempts may be made to contact them via the NDDH switchboard if time permits. There is however no guarantee that surgeons with recent experience of emergency tracheostomy will be available.

4.6 Roles must be allocated and announced at a team brief (see checklist, Appendix 2). Roles may include: team leader, 1st / 2nd intubators, assistant / ODP, sedation / drugs, monitoring and timing of attempts, runner, eFONA.

Preparation of equipment

4.7 Equipment choice is ultimately the responsibility of the airway practitioner (e.g. anaesthetist). Minimum standards for anaesthetic equipment and monitoring are set out in Association of Anaesthetists guidance.[13] In particular, capnography must be used for all attempted intubations.

4.8 Use of checklists for intubation in the emergency department or ICU should be routine. Checklists are effective in emergency / stressful situations as they reduce cognitive load and facilitate a shared mental

model. Checklists (e.g. Appendix 2) provide an aide memoire for some of the equipment that may be required. In particular, consideration should be given to the mode of pre-oxygenation including high flow nasal oxygen; type of trolley / bed; choice of laryngoscope and adjuncts; difficult airway trolley (including eFONA kit) and airway rescue equipment; and sedative administration.

4.9 Consider the use of airway ultrasound to identify and mark the cricothyroid membrane.

4.10 For most intubations the necessary equipment will be found in the airway trolleys in the anaesthetic rooms of Main Theatres, level 3 NDDH; Day Surgery Unit, level 2 NDDH; Satellite Theatres 7, 8 and Vanguard Theatre 9; in ICU; in paediatrics, and in the emergency department.

4.11 Difficult airway trolleys in each of the above areas have additional equipment including: C-MAC laryngoscopes with 3, 4 and (hyperangulated) D blades.

4.12 Paediatric difficult airway trolleys are located in Main Theatres, level 3 NDDH; Day Surgery Unit, level 2 NDDH, the emergency department and on Caroline Thorpe ward. These contain airway kit and videolaryngoscopes.

4.13 It is best to limit readily available equipment to that which will be used for planned attempts. Specialist equipment (e.g. Melker kits, for managing a cannula cricothyroidotomy) is available in column DD in Main Theatres, level 3 NDDH.

Prepare for difficulty

4.14 Airway evaluation is recommended for all patients undergoing airway interventions but is particularly important where difficulty is expected. Airway evaluation should aim to identify patients who will be difficult to intubate, patients who will be difficult to ventilate, as well as those who are at increased risk of aspiration or who will be unable to tolerate a brief apnoeic period. As the number of predictors of difficulty increases, so does the probability of actually encountering problems.

4.15 Consider review of imaging (e.g. CT scans) and nasendoscopy using an AmbuScope as part of airway assessment.

4.16 It is important to identify a strategy for airway management in advance, and this is best discussed with all members of the team at a team brief. The potential for difficulty should also be discussed with the patient. Airway management strategies may include: awake tracheal intubation, supraglottic techniques for asleep intubation, supraglottic airway insertion and invasive interventions for example cricothyroidotomy.

4.17 The airway strategy will depend on the anticipated surgery (if any), the condition of the patient (physiological reserve), patient cooperation / consent, urgency, the skills and preferences of the anaesthetist.

4.18 Consider awake intubation, plan preoxygenation (mask, CPAP, NIV or high flow nasal oxygen), plan positioning of patient (head up emphasised) and staff. Consider sedation / anaesthesia and muscle relaxation and who will be in control of these. Also consider how many attempts will be made at each stage (see DAS guidelines).

Aftercare

4.19 Ongoing management will need to be discussed with colleagues from critical care and with ENT surgeons at RD&E (Wonford). There may be a need for Level 3 transfer to another unit in which case there must be a plan for further airway difficulties en-route.

4.20 Management of critical airways is a stressful experience. It is beneficial to have a debrief after these events.

4.21 All difficult airways must be recorded in the patient notes and the problem list and the medical history updated. Consider adding the patient to the DAS difficult airway database https://das.uk.com/dad.

4.22 Can’t intubate, can’t oxygenate events, near misses and eFONA should prompt an incident report (DATIX) and the airway lead should be informed.

5. TRAINING

5.1 The additional risk posed by critical airways at NDDH can be mitigated by additional training. Airway management is teamwork and in-department training of airway teams is strongly encouraged by the Royal College of Anaesthetists and the Difficult Airway Society. This

may soon be mandated. Because of its increased exposure to risk, NDDH should be ahead of the curve in training for airway teams.

5.2 Recommendation: All those who perform emergency airway management should have access to CICO training as part of in-department airway team training.

5.3 Technical skills training should involve regular multidisciplinary out-of-theatre airway workshops, including nasendoscopy, awake tracheal intubation, airway rescue techniques for adults and children, and the use of hyperangulated blade videolaryngoscopy. Good technical skills improve intubation success and also 'make room in our heads' for good non-technical skills.

5.4 Recommendation: The department should provide an airway workshop as part of continued medical education for all airway managers. As well as technical skills training, simulation and non-technical skills training should include education on calling for help, including how to communicate the urgency with which help is required.[14]

5.5 A structured approach to the management of difficult airways is vital to ensure safety. The use of checklists and algorithms can improve situational awareness and may improve outcomes. It is recommended that the use of checklists is included in training.

6. MONITORING COMPLIANCE WITH THIS GUIDELINE

6.1 Airway incidents are discussed at anaesthetic clinical governance meetings. Incident investigation will include mapping against the requirements of this guideline to ensure compliance.

7. ASSOCIATED CLINICAL GUIDELINES OR POLICIES/PROCEDURES

7.1 National guidelines are available for the management of the difficult airway (refer to Appendix 1).

8. REFERENCES

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9. PUBLICATION DETAILS

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Appendix 1 - Live links to DAS guidelines

Unanticipated difficult intubation in adults - 2015

Critically ill adults - 2017

Extubation guidelines

Obstetric guidelines - 2015

Paediatric guidelines

Appendix 2 - DAS intubation checklist

Intubation Checklist : critically ill adults – to be done with whole team present.

Prepare the patient

- ❑ **Reliable IV / IO access**
- ❑ **Optimise position**
  - ❑ Sit-up?
  - ❑ Mattress hard
- ❑ **Airway assessment**
  - ❑ Identify cricothyroid membrane
  - ❑ Awake intubation option?
- ❑ **Optimal preoxygenation**
  - ❑ 3 mins or ETO_2 > 85%
  - ❑ Consider CPAP / NIV
  - ❑ Nasal O_2
- ❑ **Optimise patient state**
  - ❑ Fluid / pressor/ inotrope
  - ❑ Aspirate NG tube
  - ❑ Delayed sequence induction
- ❑ **Allergies?**
  - ❑ ↑ Potassium risk?
    - avoid suxamethonium

Prepare the equipment

- ❑ **Apply monitors**
  - ❑ SpO_2 / waveform $ETCO_2$ / ECG / BP
- ❑ **Check equipment**
  - ❑ Tracheal tubes x 2
    - cuffs checked
  - ❑ Direct laryngoscopes x 2
  - ❑ Videolaryngoscope
  - ❑ Bougie / stylet
  - ❑ Working suction
  - ❑ Supraglottic airways
  - ❑ Guedel / nasal airways
  - ❑ Flexible scope / Aintree
  - ❑ FONA set
- ❑ **Check drugs**
  - ❑ Consider ketamine
  - ❑ Relaxant
  - ❑ Pressor / inotrope
  - ❑ Maintenance sedation

Prepare the team

- ❑ **Allocate roles**

One person may have more than one role.

  - ❑ Team Leader
  - ❑ 1st Intubator
  - ❑ 2nd Intubator
  - ❑ Cricoid force
  - ❑ Intubator's assistant
  - ❑ Drugs
  - ❑ Monitoring patient
  - ❑ Runner
  - ❑ MILS (if indicated)
  - ❑ Who will perform FONA?
- ❑ **Who do we call for help?**
- ❑ **Who is noting the time?**

Prepare for difficulty

- ❑ **Can we wake the patient if intubation fails?**
- ❑ **Verbalise** "Airway Plan is:"
  - ❑ **Plan A:**
    Drugs & laryngoscopy
  - ❑ **Plan B/C:**
    Supraglottic airway
    Face-mask
    Fibreoptic intubation via supraglottic airway
  - ❑ **Plan D:**
    FONA
    Scalpel-bougie-tube
- ❑ **Does anyone have questions or concerns?**